Quantum Simulation of Collective Proton Tunneling in Hexagonal Ice Crystals

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The effect of proton tunneling on many-body correlated proton transfer in hexagonal ice is investigated by quantum simulation. Classical single-particle hopping along individual hydrogen bonds leads to charge defects at high temperature, whereas six protons in ringlike topologies can move concertedly as a delocalized quasiparticle via collective tunneling at low temperature, thus preventing the creation of high-energy topological defects. Our findings rationalize many-body quantum tunneling in hydrogen-bonded networks and suggest that this phenomenon might be more widespread than previously thought.

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Proton tunneling and zero-point motion govern many important hydrogen(H)-bonded systems such as water nanostructures on metal surfaces [1], various ice phases [2,3], and elementary biosystems [4,5]. Because these effects strongly influence the strength of H bonds [6] and sensitively depend on the detailed structure and potential energy surface, ice phases often serve as model systems for more complex H-bonded systems such as water and aqueous solutions [7–10]. For example, cubic ices undergo phase transitions induced by quantum delocalization and incoherent single-particle tunneling at Mbar pressures [3,11]. Ordinary (hexagonal) ice I_h displays many peculiar phenomena, which are unrelated to anomalies in supercooled water or amorphous ice, such as anomalous lattice expansion upon $H \rightarrow D$ substitution [12] and widely varying surface vacancy formation energies reminiscent of amorphous materials [13]. Protons in ice I_h occupy one of two sites along an H bond subject to the constraint of leaving all water molecules intact [14]. This proton disorder leads to residual entropy [14] and renders ice I_h a prototypical frustrated material, serving as a model for spin ices [15,16] and H-bonded ferroelectric materials exhibiting proton tunneling [17–19].

In contrast to the above-mentioned, incoherent single-particle tunneling, coherent many-body tunneling in condensed-matter systems has remained scarcely investigated. Exceptions include double proton transfer [20,21] or concerted transfer of at most four protons in molecular crystals [22,23]. A recent neutron scattering study on ice I_h [24], however, revealed an unexpected temperature dependence and H/D isotopic substitution effect, which have been interpreted schematically in terms of collective tunneling of six protons within locally proton-ordered hexagonal rings [24] (see inset of Fig. 4 therein). These findings are puzzling because the proton-disordered ground state of ice I_h is considered to prevent such efficient proton motion in the absence of doping-induced excess defects [14]. Further, in contrast to well-understood single-proton tunneling along individual H bonds, our mechanistic insights into collective tunneling of many protons in complex H-bonded networks, including ices [25], still remain elusive.

In this Letter, we employ fully quantum-mechanical *ab initio* path integral simulations [26–28], which provide reliable descriptions of realistic H-bonded materials and overcome the limitations of model studies based on empirical double-well potentials [29], to show that concerted proton tunneling of six protons can occur in low-temperature ice I_h. Importantly, a comparison between the low-temperature quantum regime and the high-temperature classical regime reveals qualitative differences in the transition mechanism. At high temperature, decoupled single-proton hopping along individual H bonds creates charge defects, causing a large free energy barrier. At low temperature, collective proton tunneling in cyclic topologies via a delocalized quasiparticle largely circumvents topological defects and the associated energy penalty, thus underlining the importance of tunneling for the collective transfer of six protons.

Our model of proton-disordered ordinary ice was defect free [14], exhibited zero dipole moment [30], and contained one proton-ordered hexagonal ring (cf. the insets of Fig. 1). The proton motion within this proton-ordered six ring comprised single-particle transfer along individual H bonds and collective transfer of all six protons whose progress was measured by the local and global collective variables (CV) $\phi_i = d(D_iH_i) - d(A_iH_i)$ and $\Phi = \sum_{i=1}^{6} \phi_i/6$, respectively. Above, $d(D_iH_i)$ and $d(A_iH_i)$ are the distances of the proton to the donor and acceptor oxygen sites in the initial state of the ith H bond, $D - H \cdots A$.

Investigating the effect of tunneling and zero-point motion on correlated proton transfer requires techniques treating both electrons and nuclei quantum mechanically. Thus, we described the electronic structure via density functional theory [31] and compared classical and quantum nuclear motion via *ab initio* molecular dynamics and *ab initio* path integral simulations [28]. In the quantum simulations, cyclic discretized path integrals rigorously represent each nucleus as a necklace consisting of

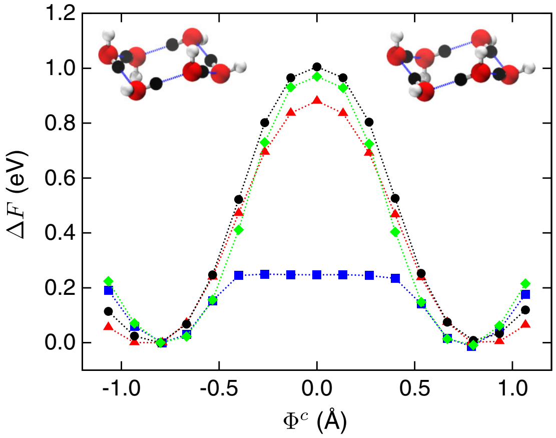

FIG. 1 (color online). Free energies as a function of the global collective variable Φ^c from quantum (blue squares: 50 K; red triangles: 320 K) and classical (black circles: 300 K, green diamonds: 50 K) simulations; linear dotted lines are guides to the eye. The left and right insets schematically show the proton configuration of the ordered six ring in the initial ($\Phi^c = -0.8$ Å) and final ($\Phi^c = +0.8$ Å) states, respectively. Note that these states are quasidegenerate in view of both the cyclic character of the six rings and insignificant influence of the proton-disordered environment in stark contrast to noncyclic topologies. H bonds, oxygen atoms, and hydrogen atoms are shown by thin lines, large spheres, and small spheres, respectively.

$P \to \infty$ beads, which are connected via harmonic springs. At low temperature and low mass, quantum fluctuations cause delocalization because beads deviate from their center of mass (centroid), thus representing the spread of wave functions, whereas all beads coincide with their centroid at high temperature and high mass. Because the maximum delocalization of the proton (~0.5 Å, see Supplemental Material, Fig. S1 [31]) was smaller than the smallest relevant H-H distance (~2.3 Å), nuclear exchange could be neglected. Moreover, the neutron scattering signal [24] is insensitive to whether the scattering particle is a boson or a fermion.

As expected, neither proton transfer nor proton tunneling occurred spontaneously for our model, thus requiring appropriate rare-event techniques. Specifically, free energy differences were computed via thermodynamic integration [32], which has routinely been applied to molecular dynamics [33] and path integral simulations [34] to sample configurations of low probability. Herein, we constrained the global CV centroid Φ^c, thus generalizing that for single-proton tunneling [35], at various values of Φ^c to systematically enforce the transition from the initial to the final state. For each value of Φ^c, a different region of phase space was explored, and the resulting constrained averages were integrated to reconstruct the free energy profile (see Supplemental Material, Tables S1 and S2 [31]).

The quantum and classical free energy profiles along the global CV, Φ, displayed in Fig. 1 for selected temperatures, reveal a "classical" and a "quantum" regime. In the former, the bell-shaped free energy barrier of ≈1 eV is similar at 300 and 50 K and thus virtually identical to the potential energy barrier of the process. Such high barriers prevent thermally activated transitions from the initial to the final ordered state via transfer of all protons along their H bonds, as depicted in the insets of Fig. 1. We have also checked that both different proton-disorder configurations outside the proton-ordered six ring in the hexagonal plane and proton-ordered six rings perpendicular to that plane only lead to small quantitative differences, whereas the qualitative scenario discussed herein is retained. Moreover, the barriers from quantum and classical simulations at ~300 K show that thermal fluctuations dominate the classical regime, whereas zero-point motion only causes minor corrections of ~10%.

In contrast, low-temperature quantum simulations at 50 K yield a substantially lower activation free energy. Although the activation barrier, as computed herein, cannot directly be related to the quantum rate constant [36], the different shape of the free energy profile indicates qualitatively relevant quantum fluctuations, which will prove critical to the understanding of the unveiled quantum phenomenon. Having confirmed the phenomenon as such, in accord with experimental evidence [24], we next turn to its detailed mechanistic analysis.

To quantify proton delocalization, a single-particle indicator of quantum fluctuations, we computed the radius of gyration of all protons in the ordered six ring whose distribution is shown in Fig. S1 [31]. As expected, the mean radius of gyration at low temperature exceeds that at high temperature in both the transition and initial state, and, in the latter, the distribution coincides with that of the remaining protons in the system (see Fig. S1 [31]). At high temperature, the distribution is virtually identical in the initial and transition state. In contrast, the quantum delocalization substantially increases in the low-temperature regime when moving from the initial to the transition state, already indicating extreme quantum fluctuations at 50 K at the level of individual protons.

Transcending such a single-particle perspective, we analyzed the imaginary-time correlation function of the global CV, $C(\tau) = \langle|\Phi(\tau) - \Phi(0)|^2\rangle^{1/2}$, thus involving all six topologically connected protons and measuring fluctuations within a quantum path. At high temperature, the flat and confined $C(\tau)$, shown in Fig. 2, shows localized motion of all six protons along the transition. In contrast, the nearly parabolic shape at the low-temperature transition state demonstrates [35,37] delocalized quantum motion due to tunneling that involves collectively six protons, suggesting that the global CV Φ describes a quasiparticle comprising all protons in ordered six rings behaving collectively in the sense of tunneling.

Having revealed the distinct quantum nature of the proton motion at low temperature, we further elucidate its collective character. The quasiparticle nature of the six

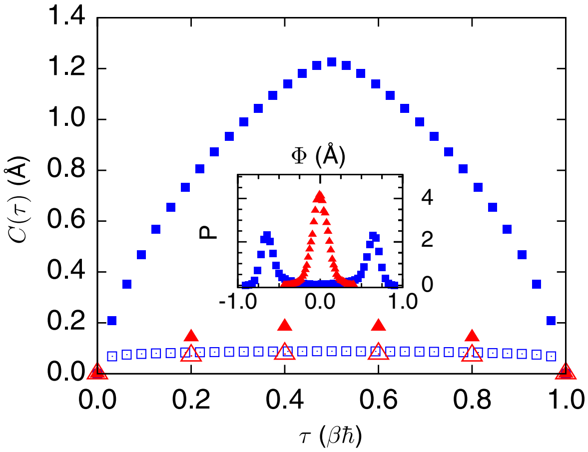

FIG. 2 (color online). Imaginary-time correlation functions of the global collective variable $C(\tau)$ from quantum simulations (squares: 50 K; triangles: 320 K) in the initial ($\Phi^c = -0.8$ Å; open symbols) and transition ($\Phi^c = 0$ Å; filled symbols) state. The inset shows corresponding global collective variable distributions $P(\Phi)$ at the transition state (squares: 50 K; triangles: 320 K).

protons within proton-ordered six rings is indicated by the distribution $P(\Phi)$ of the global CV, shown in the inset of Fig. 2, at the transition state, where $\Phi^c = 0$ Å yields $\langle\Phi\rangle = 0$ Å. In the high-temperature limit, this distribution is unimodal and peaked around zero. The statistical analysis of the trajectory at $\Phi^c = 0$ Å reveals that three proton transfers have already and three not yet occurred in the majority of configurations, thereby creating at least one topological defect pair on average, as shown in Figs. S2 and S3 and Videos S1 and S2 [31]. The resulting charge separation in terms of local H_3O^+ and OH^- high-energy defects causes the large free energy barrier in the classical regime (cf. Fig. 1).

At variance with this defect-mediated proton-transfer scenario, the low-temperature quantum transition state exhibits a bimodal distribution peaked at Φ values corresponding to the initial and final states, thereby also leading to a vanishing operator expectation value: $\langle\Phi\rangle = 0$ Å. Hence, the number of topological defects is extremely low in the quantum case at 50 K, thus lowering the free energy barrier considerably compared to the classical behavior at 50 K. In-depth statistical analysis, see Fig. S2, reveals a unimodal distribution peaked at three transferred protons in the classical regime and a sharp bimodal distribution at zero and six transferred protons in the quantum case, thus essentially fully avoiding charge defects. This behavior strongly supports the picture of a quasiparticle described by Φ and composed of all six protons that move in a fully correlated manner from the initial to the final state via a transition state characterized by deep quantum tunneling.

This absence of topological defects and the correlated transfer of all six protons even at the transition state are necessary conditions for collective proton tunneling in the ordered six ring. However, the final proof of concerted tunneling requires the transition from the initial to the final state within Feynman paths, characterized by kink-antikink pairs along imaginary time τ. The resulting instantons are very short-lived in imaginary time, as convincingly visualized in Fig. 3 by a plot of $\Phi(\tau)$ representative of the statistical ensemble and characteristic of the deep tunneling regime. The fully synchronized imaginary-time evolution of the global CV Φ with that of all six local CVs $\{\phi_i\}$ implies perfectly correlated motion within the proton-ordered six ring and thus uniquely demonstrates tunneling of the quasiparticle as an entity. Beyond just inspecting individual snapshots, we computed the average difference $\Delta(\tau, t_k) = (\sum_{i=1}^{6} |\Phi(\tau, t_k) - \phi_i(\tau, t_k)|^2/6)^{1/2}$ between the global and six local CVs, i.e., deviations between $\Phi(\tau)$ and all $\phi_i(\tau)$ at each sampled configuration t_k and imaginary time τ. Thus, $\langle\Delta\rangle \equiv 0$ Å if all local CVs always exactly follow $\Phi(\tau)$. For the initial state, the distribution $P(\Delta)$ is sharply peaked at a small value Δ_{max} at low temperature (cf. the inset of Fig. 3 or Fig. S7 [31]), which quantifies ordinary quantum broadening at 50 K due to zero-point motion. Highly surprisingly, its shape at the transition state at 50 K is statistically indistinguishable from that of the initial state, thereby unveiling that all six protons must move collectively as a quasiparticle at low temperatures, as visualized in Video S3 [31], recorded at the transition state $\Phi^c = 0$ Å.

In contrast to the quantum case, Δ_{max} of $P(\Delta)$ at the transition state is twice that of the initial state at

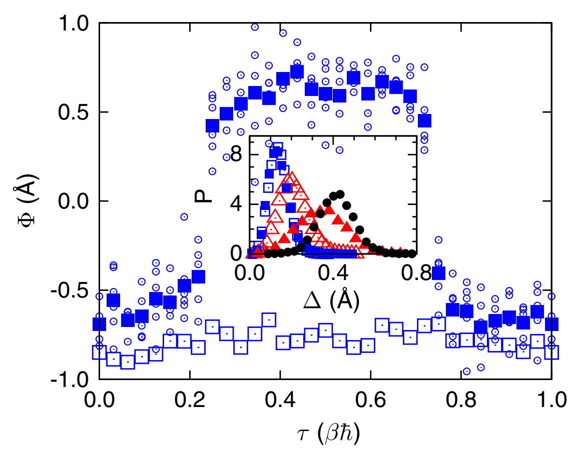

FIG. 3 (color online). Representative world-line plot of the global collective variable $\Phi(\tau)$ for typical initial (large open squares) and transition (large filled squares) state configurations from quantum simulations at 50 K. The superimposed small open circles show the corresponding world-line plots of all six local collective variables $\{\phi_i\}$ at the transition-state configuration. The inset (see also Fig. S7 [31]) shows the distribution $P(\Delta)$ of the measure for deviations between the global and local collective variable from quantum (squares: 50 K; triangles: 320 K) and classical (circles: 300 K) simulations in the initial ($\Phi^c = -0.8$ Å; open symbols) and transition ($\Phi^c = 0$ Å; filled symbols) state.

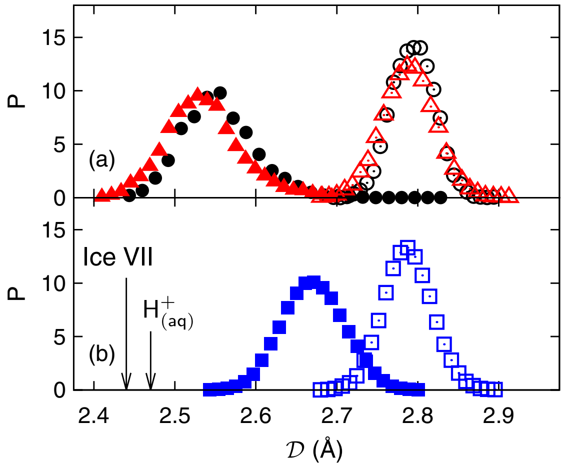

FIG. 4 (color online). Distribution of the global donor-acceptor O-O distance in the six ring $\mathcal{D}$ from quantum (squares: 50 K; triangles: 320 K) and classical (circles: 300 K) simulations in the initial ($\Phi^c = -0.8$ Å; open symbols) and transition ($\Phi^c = 0$ Å; filled symbols) state; all high-temperature data are collected in the upper panel. For comparison, arrows mark the average contracted O-O distances for incoherent single-proton tunneling [3] in ice VII, ≈2.44 Å at 25 K [38], and for single-proton transfer in bulk water [7], ≈2.47 Å at 300 K.

320 K. This high-temperature behavior arises from the noncollective single-particle transfer of individual protons along their H bonds, thus accumulating large instantaneous deviations $|\Phi(\tau) - \phi_i(\tau)|$ in line with the aforementioned high defect concentration. This detailed analysis of imaginary-time correlation and distribution functions together with the observed temperature dependence clearly demonstrates that concerted motion of six protons, i.e., quasiparticle tunneling, within ordered six rings occurs at low temperature, whereas defect creation characterizes the high-temperature regime.

Finally, according to conventional wisdom, proton tunneling in ice I_h should be prevented at ambient pressure because of the large O-O distances, $d \sim 2.8$ Å, and thus high barriers. Indeed, the defect-mediated single-proton transfer in ice I_h at high temperature requires strong contraction of the O-O distance when moving from the initial to the transition state, as evidenced by the global donor-acceptor distance, $\mathcal{D} = \sum_{i=1}^{6} d(D_i A_i)/6$, shown in Fig. 4. At high temperature, the quantum and classical scenarios essentially coincide, thus supporting the thermal nature of the process. In contrast, the low-temperature quantum scenario reveals a surprisingly moderate O-O contraction upon reaching the transition state in view of quasiparticle tunneling. Such behavior is unknown from earlier quantum simulations on the pressure-induced quantum transition of ice VIII to VII due to incoherent, defect-mediated single-proton tunneling at 25 K [3] and single-proton transfer along individual H bonds in bulk water at 300 K [7], which require significant O-O contractions to $d \approx 2.44$ Å and to $d \approx 2.47$ Å, respectively. Our observation of collective proton tunneling in ordinary ice at ambient pressure, extracted from the analysis of sampled quantum paths, thus challenges previous claims according to which large donor-acceptor distances prevent tunneling in ice I_h.

In summary, our *ab initio* path integral simulations show that quantum fluctuations in ordinary ice I_h at ambient pressure allow for collective tunneling of six protons in proton-ordered six rings at low temperature. This correlated tunneling process prevents the creation of energetically costly topological defects and requires no strong shrinking of the donor-acceptor distances within the heavy-atom skeleton because all six protons tunnel concertedly, and not individually in an uncorrelated fashion, as in the well-studied ice VIII to VII transition. The low-temperature quantum behavior differs qualitatively from the thermally activated hopping along individual H bonds at high temperatures, which leads to ionic defects causing a large activation barrier. The emerging picture in the quantum regime consists of a quasiparticle comprising six protons moving in a fully correlated manner from the initial to the final state via a transition state characteristic of deep collective quantum tunneling.

This phenomenon might be an instance of the sought-after coherent tunneling state involving many protons directly connected by wirelike H bonds in the condensed phase. The present findings thus provide fresh insights into the correlated quantum motion of coupled protons in H-bonded systems embedded in fluctuating, symmetry-breaking environments and suggest that many-body tunneling might be important in a variety of systems, including a wealth of other ice phases and, most noteworthily, cyclic H-bonded water clusters on metal surfaces.

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